

CALIBRATION REQUIREMENTS OF AN AUTOMOTIVE HUD INTERFACE USING A VIRTUAL ENVIRONMENT: METHODOLOGY AND IMPLEMENTATION

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ABSTRACT

This paper presents the methodology of a system which utilises a virtual reality (VR) environment to establish the calibration requirements for a prototype automotive full-windscreen Head-Up Display (HUD) interface. An easy and fast calibration of the HUD can be a considerable advantage for the final product since rapid customisation is crucial in everyday driving, especially if the same vehicle is often being used by more than one driver. Simulating the positioning of the HUD, in accordance to the driver, in a VR environment is a cost-effective way to identify design limitations before proceeding towards physical mock-ups. The system was designed to enable the evaluation of additional usability aspects such as testing the effectiveness of the HUD interface under varying visibility conditions.

KEY WORDS

VR simulation, methodology, 3D calibration, HUD, automotive engineering

1. Introduction

Head-Up Display (HUD) systems need to be calibrated in accordance to the driver's body or viewing position before their use, like any other component which requires optimal positioning in order to function properly (i.e. side mirrors). The calibration of such components can be achieved either manually or mechanically (e.g. hydraulic, electric actuators) and can be either adjusted by human intervention or automatically. Normally, the calibration requirements of such systems depend on the usage context and the interaction between multiple factors of the system (i.e. projection method, haptic interface etc.) [1]. Although generic design recommendations tend to be as precise as possible, the literature suggests that in particular cases they can be ambiguous given that technological advances are constantly creating new

opportunities for various design and evaluation processes. The interface under investigation falls in such a category as automotive HUD interfaces have not been extensively studied or utilised thus far, especially those applied on a large segment of the windshield. Hence, this study addresses the trade-off between the suggestions and demands of different design guidelines in terms of operational requirements. As highlighted in the HARDIE guidelines, these guiding principles do not pose any constraints for development of possible alternative methods, which in this case were essential for the evaluation process of the prototype. Since we entertain the possibilities to fully integrate the HUD interface in a contemporary vehicle in the forthcoming year, we opted to identify possible problems and limitations, and to estimate calibration requirements using a VR mock-up.

In this paper we present the first of the three stages which need to be fulfilled prior to the implementation of a mechanical/human-operated calibration of the HUD system. The three stages namely are: calibration requirement analysis based on a VR simulation environment, system evaluation using a driving simulator and, finally, physical system implementation on a real vehicle.

The first stage involves the construction of a realistic 3D model of a vehicle's interior that serves as a test-bed for different calibration precision levels and techniques. Apart from a detailed representation of a three-dimensional interior, the VR environment has to provide an identification of the calibration hierarchy and a VR representation and experimental calibration of the symbols in relation to the driver's physique and field of view. As the actual hardware implementation of the interface is under development, the calibration of the system could only occur as a simulation in a virtual environment. The rationale behind such an attempt is to identify the potential difficulties and problems that could occur during the construction of this system by initially visualising and testing it in a VR environment.

2. HUD Overview

Provided that driving is a task performed using primarily our sense of sight [3], various studies have explored the use of visual signals to communicate useful information to the driver [4, 5, 6], whereas other avenues (i.e. auditory signals) have also been investigated. The literature outlines the advantages of HUD systems for such purposes compared to Head-Down Displays (HDD) [4, 6, 7]. The majority of existing HUD designs falls into the small projection category that exploits only a small part on the windshield, usually within the driver's field of view. Too often though, the wealth of information provided via a small HUD obscures the real-life environment and confuses the driver due to visual clutter. Therefore, in this design, we decided to adopt the full-windshield HUD option which utilises a large portion of the windshield area as opposed to the small projection form described above. Research by [4, 6] reports that the full-windshield form can be a valid alternative to small HUDs with comparable performance gains in contrast to classic HDD instrumentation.



Figure 1: Screenshot of original HUD design

The proposed HUD [8] has been designed for use under low visibility conditions, such as fog and heavy rain, in motorway environments. The projected graphical symbols (a series of size-coded and colour-coded icons) provide information about the surrounding exterior environment (i.e. neighbouring vehicles, barriers, hard shoulder lane). Thus, considering the nature and the format of the information it was deemed more suitable to use a full form design as it is shown in Figure 1. During the development of the HUD display, four pieces of information were primarily identified as the most crucial for collision avoidance in motorways. This information was visualised through iconic representation and produced four symbols, namely lane recognition, lead vehicle detection, traffic warning and sharp turn notification symbols as presented in Figure 2. Prior to commencing the design and modelling of these four

symbols, valuable comments had been gathered from drivers during informal interviews.

To achieve the goal of designing attention-seeking symbols, it was decided to create colourful visual cues adhering to the SAE colour coding standards. The symbols would also be able to change dimensions or visual intensity through automatic or manual contrast and brightness adjustments. Particularly, depending on the situation, a number of effects have been developed for each symbol. The primary challenge was to provide auxiliary information to the driver without diverting significant mindshare from the primary task. At the same time, the visual cueing effects may have ended up being subtle enough to go unnoticed. The effectiveness of the icon system has been verified through extensive user testing as presented in [8].

3. Interface Initial Positioning

Earlier work by Yoo et al [6] has studied the full-windshield HUD regions with respect to response times (RTs) by the driver on symbol appearance. The relevant regions are depicted in Figure 2. According to the results in [5] the sections that provided the best RTs were the ones forming the second row in Figure 2, as they were located in the horizon line in front of the driver's field of view. The first and third row also gave a few equally low RT measurements. By following the grid pattern in [6], we positioned the lead-vehicle symbols in the fastest RT sections. The lane symbols were projected in C3 and E3, which offer moderate RTs, and the turn symbol was placed within the D3 square just below the middle lead-vehicle. Since the turn symbol's function is to provide early warning, it was positioned in a slightly slower RT area which seemed more appropriate and non-distracting. During the evaluation of the HUD interface on a custom-made driving simulator that had been designed specifically to test the interface's abilities, we verified Yoo et al's position and results regarding the RT in the charted areas of the windscreen [6]. Consequently we incorporated an initial generic calibration layer and positioned the symbols on a large scale grid of 5x3 (Figure 2), which corresponds to the part of the windshield through which the driver observes the traffic and environment. A detailed analysis of the methodology used to support the calibration function is provided further on.

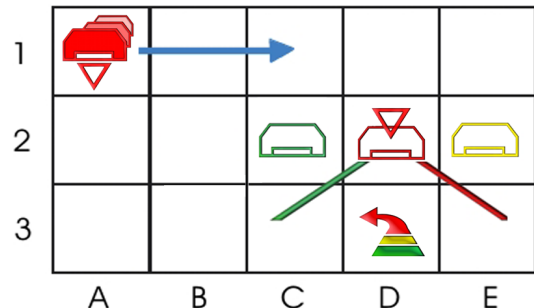


Figure 2: Design of the HUD layout

4. Methodology of Layers

The calibration of the interface incorporates a layered hierarchy of projected screens, which defines a increasingly precise alignment with the user's field of view. These screens are projected on top of a simulated background which has been reconstructed from the existing driving simulator data used for the evaluation of the HUD icons. The components that comprised the VR calibrating system were developed in a manner that would assist faster interaction between the user and the interface, therefore minimising the time required for the calibration process. The following sections will provide information for the proposed set-up in the order that the layers have been applied in the VR environment, as depicted in Figure 3.

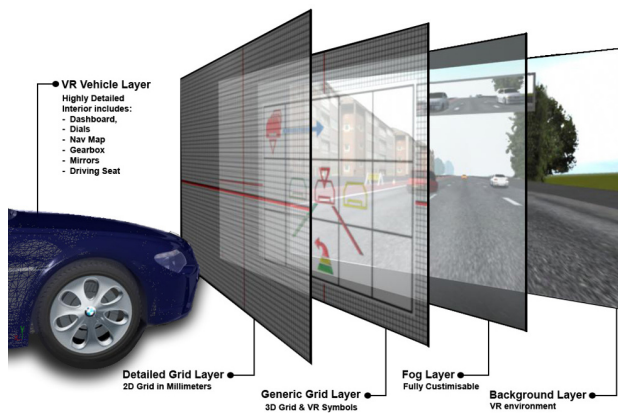


Figure 3: The different calibration layers

4.1 Virtual Motorway Layer

A VR depiction of the driving simulator environment was an ideal solution for providing the method with a well developed and tested background. The realistic visualisation of vehicles and other objects (i.e. buildings, bridges, tarmac and barriers) that complement the surroundings proved to have a considerable effect on the driver's spatial recognition and perception as it is described in [8].

4.2 Fog Layer

The primary goal of the HUD system is to enhance the driver's senses and spatial awareness under low visibility conditions, hence the incorporation of a fog layer was considered essential. Furthermore, the brightness and contrast of the symbols can be altered to improve the effectiveness of the appearing symbols according to the driver's physiology and contrast preferences. The driver can also adjust the saturation of the colours up to a certain degree so as not to corrupt the primary colour coding as suggested by ISO and SAE [10]. The fog is simulated by the graphics hardware as an exponential fade to grey depending on the distance of an object.

4.3 Generic Grid

This layer is the first part of the calibration as it shows the symbol's default positions following Yoo et al's pattern of appearance, which defined the faster response squares. This pattern is enlarged to fill the complete driver's field of view on the windshield, therefore maximizing the usable HUD area. The grid suggests the main positioning of the symbols in relation to a highlighted horizon line. This improvement in the original positioning grid [6] supports mostly the calibration process in respect to the horizon line and driver's field of view.

Positioning the generic grid allows us to establish the extents to which the HUD can be calibrated while still maintaining the basic desired structure.

4.4 Detailed Grid

This layer represents in a Cartesian diagram a grid formed in the millimetre scale. This level of calibration will be used for refining the positioning acquired by the *generic positioning grid*. The driver can easily transfer the symbols (e.g. turn or traffic warning symbols) in neighbouring positions a few millimetres away from the original predefined optimal position.

4.5 Vehicle Interior Layer

For the purpose of the experiment we modelled the interior of a BMW 6 series in high detail for a precise, realistic simulation while keeping the model complexity (polygon count) within reasonable limits for real-time visualisation. We opted for that particular vehicle because it has been already equipped with a small form HUD interface. It was therefore feasible to examine the possibility of the application of a larger version HUD such as the proposed one.

By combining the pre-mentioned layer structure into one sandwich-projection approach (Figure 3) we can provide the driver with a realistic representation of the calibrating process of the proposed interface while at the same time experiment with the symbols' design, variation and projection position as well as with the projected environment, the traffic and the weather conditions simulated.

5. VR Implementation

Using a virtual environment enables us to simulate a large range of configuration options without the need to build a physical mock-up. Based on the model of the car interior and a motorway environment, we can simulate the effects of different icon types, positioning, and global calibration parameters. Using a spatially immersive, stereoscopic display enables a simulation at 1:1 scale including effects of depth perception.

We chose Multigen's VEGA Prime visual simulation environment as the software development platform for the

simulator. Besides standard VR software toolkit functionality, such as stereoscopic rendering and device interfacing, VEGA Prime offers realistic simulations of the environment including time of day, weather and visibility effects such as fog, snow and rain.

The display hardware consists of a back-projection surface, a Barco CRT active stereo projector driven by a dual processor PC with a high-end graphics card (nVidia Quadro FX 4400). No special input devices were required for the virtual mock-up.



Figure 4: User in the simulator with stereoscopic display

5.1 Environment Simulation

The simulator provides a minimal environment required to test the HUD. It simulates driving on a straight, infinitely long motorway. Additional cars with basic driving logic including randomized speed, overtaking simulation and basic collision avoidance logic provide a busy traffic situation. The interior of the car is modelled in high detail to provide a realistic feeling of sitting inside a real car.

The system simulates various weather conditions to show the effectiveness of the HUD under varying visibility. The fog effect can be set anywhere from a clear day to only a few meters visibility (Figure 5). A time of day setting provides control over light intensity, colour cast and angle. Rain and snow provides additional obstruction, although their effect is negligible unless combined with fog simulation.



Figure 5: Simulation of environmental effects a) Clear day b) Fog with rain c) Night

The environment and the interior of the car are modelled in a 1:1 scale. In combination with a precisely calibrated large stereoscopic display system, this provides a correct impression of the size and distance.

Head-tracking of the user in front of the display is not used in this simulation, but it could be added with little effort. Assuming a static head position is a reasonable assumption in a car-driving scenario of this kind.

5.1 Icon Positioning

It is assumed that the HUD system always knows the spatial position of the cars within its vicinity based on radar, GPS and dynamic networking information as discussed in [11].

The system renders the HUD icons on a virtual flat plane parallel to the windshield. The position of an icon is determined by the intersection of the ray between the head position and the actual position of the car. The size of the icon is determined by the ratio between the distance between the head and the display and the distance between the head and the car, which corresponds to the actual visible size of the car in front [12]. A global scale factor can be applied to adjust the icon size to the preference of the user, providing a trade-off between intrusiveness vs visibility.

The icon rendering plane is placed at the exact location of the windshield in the current set of tests, simulating a HUD system that directly projects onto the glass [13, 14]. The plane can be moved further forward or backward, simulating the effect of more sophisticated display systems that enables a focus plane further away.

5.2 Head Position and Calibration

The virtual viewer head can be positioned freely within the car environment, simulating the different seating preferences and body height of different drivers [15, 16, 17]. As the icon placement depends on the head position, varying the viewpoint clearly demonstrates the limits of the usable display area depending on the seating. Figure 6 shows the configuration for various body heights.



Figure 6: Driver's perspective a) Typical driver b) Tall driver c) Short Driver

The head position used for the icon position calculation can be displaced arbitrarily from the real viewing position to simulate calibration errors. As shown in Figure 7, even a small horizontal and vertical offset of 2 cm can result in a significant offset of the icon, reinforcing the need for precise calibration or even head-tracking.

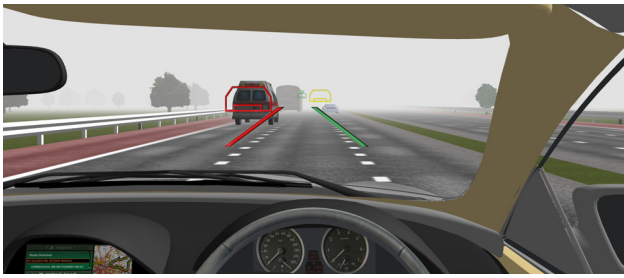


Figure 7: Calibration error: the viewpoint for the icons is offset by 2cm both vertically and horizontally

6. Discussion

This paper presented a methodology to design and adjust a car head-up display using the full windshield area. While previous experiments using a car simulator application with a more basic 2D projection environment proved the general suitability of the visual language and icon selection [6], the new system now enables us to test the ergonomics of the proposed solution much more accurately.

We found that Yoo's model of placing icons into areas that strongly attract attention nicely and naturally corresponds to the direct overlay of distant cars with the corresponding icons for a reasonable range of driver's body heights and seating position, as shown in the examples in Figure 6. Unusually tall or short drivers would require the display of symbols in ineffective areas of the user's field of view or even on areas hidden by the dashboard or the car roof, unless the driver's seat and steering wheel provides a sufficient adjustment range.

Using a 1:1 scale model and projection of the car, HUD and the environment enables us to test the ergonomics relating to the field of view and the impact of visual clutter and attention attraction in a realistic manner, whereas previous tests only allowed indirect conclusions and might have been influenced by the relatively narrow field of view imposed by using a standard wide screen TV for the visual stimulus.

While the majority of placement evaluation and error analysis can be done using our 3D model but with 2D projection only, the stereoscopic projection provides a much deeper insight into the limitations of the system. Even though the HUD system is clearly designed for low visibility conditions where the driver relies on the HUD information almost exclusively, a naïve 2D simulation might mislead the designer into thinking that the system could provide useful information on a clear day for distant objects. The stereoscopic simulation, however, clearly reveals the difficulties induced by the different focusing planes of the HUD icon vs. the actual car. This is less apparent in very low visibility situations where the distance between the farthest visible object and the HUD plane is less pronounced, and therefore the added distraction by the HUD is offset by the additional information that would otherwise not be available. Figures 8 and 9 show the alignment error introduced by

the distance between the eyes for the left and right eye respectively under different visibility conditions, with cars far away losing the correspondence almost completely. Future versions our environment will enable to simulate the effect of more enhanced HUD technology that shifts the focus distance further out through using special optics.

Additional effort will need to go into the simulation of light effects, such as glare on the windshield on a sunny day or from headlights of cars driving into the opposite direction, or reflections in rain particularly during night time. The latest advances in real-time simulation technology, particularly special effects calculated on modern programmable graphics hardware, will allow us to push the simulation fidelity even further.

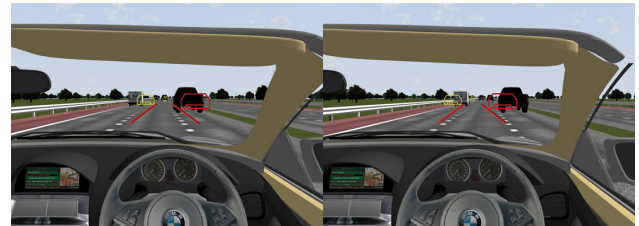


Figure 8: Alignment problems due to different focal plane revealed through stereoscopic rendering. Left/right eye view



Figure 9: Alignment problems are much less apparent under low-visibility conditions and thus less distracting. Left/right eye view

7. Conclusions

This study presented an innovative method for engineering visualisation and evaluation that provides the ability to alter the display parameters of a specific type of notification system and calibrate it to conform to the user's physique. The initial testing of this method highlighted the importance of the consideration of the interface's characteristics in conjunction to their three dimensional positioning. The VR environment simulated convincingly the driving simulator environment, creating consistent visual cues which could be identified by the drivers in both simulation environments, thus minimising the familiarisation trial period before every experiment. Our future work will entertain the possibility of changing the interface components and adjusting the position not only for different drivers but also for different vehicles. This would provide our research with information

concerning the compatibility of the proposed HUD interface with a larger variety of vehicles. Finally it is in our future plans to verify the results of this method in more realistic driving situations with the use of our prototype HUD system.

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